KISS-ICP: In Defense of Point-to-Point ICP – Simple, Accurate, and Robust Registration If Done the Right Way

Ignacio Vizzo Tiziano Guadagnino Benedikt Mersch Louis Wiesmann Jens Behley Cyrill Stachniss

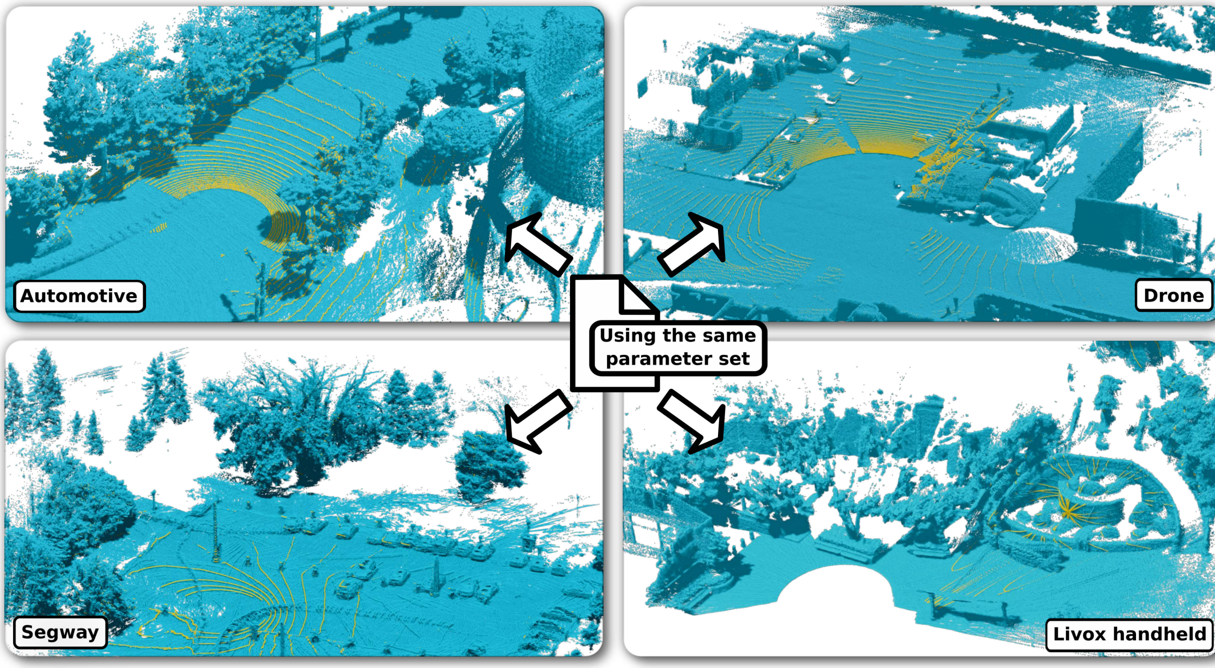

Fig. 1: Point cloud maps (blue) generated by our proposed odometry pipeline on different datasets with the same set of parameters. We depict the latest scan in yellow. The scans are recorded using different sensors with different point densities, different orientations, and different shooting patterns. The automotive example stems from the MulRan dataset [15]. The drone of the Voxgraph dataset [23] and the segway robot used in the NCLT dataset [5] show a high acceleration motion profile. The handheld Livox LiDAR [17] has a completely different shooting pattern than the commonly used rotating mechanical LiDAR.

***Abstract*—Robust and accurate pose estimation of a robotic platform, so-called sensor-based odometry, is an essential part of many robotic applications. While many sensor odometry systems made progress by adding more complexity to the ego-motion estimation process, we move in the opposite direction. By removing a majority of parts and focusing on the core elements, we obtain a surprisingly effective system that is simple to realize and can operate under various environmental conditions using different LiDAR sensors. Our odometry estimation approach relies on point-to-point ICP combined with adaptive thresholding for correspondence matching, a robust kernel, a simple but widely applicable motion compensation approach, and a point cloud subsampling strategy. This yields a system with only a few parameters that in most cases do not even have to be tuned to a specific LiDAR sensor. Our system performs on par with state-of-the-art methods under various operating conditions using different platforms using the same parameters: automotive platforms, UAV-based operation, vehicles like segways, or handheld LiDARs. We do not require integrating IMU data and solely rely on 3D point clouds obtained from a wide range of 3D LiDAR sensors, thus, enabling a broad spectrum of different applications and operating conditions. Our open-source system operates faster than the sensor frame rate in all presented datasets and is designed for real-world scenarios.**

***Index Terms*—Mapping; Localization; SLAM**

I. Introduction

ODOMETRY estimation is an essential building block for any mobile robot that needs to autonomously navigate in unknown environments. In the LiDAR sensing domain, current odometry pipelines typically use some form of iterative closest point (ICP) to estimate poses incrementally [10], [26], [31], [35]. Even though LiDAR odometry has been an active area of research for the last three decades, the design of current systems is usually coupled with assumptions about the robot motion [10] and the structure of the environment [28] to

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All authors are with the University of Bonn, Germany. Cyrill Stachniss is additionally with the Department of Engineering Science at the University of Oxford, UK, and with the Lamarr Institute for Machine Learning and Artificial Intelligence, Germany.

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KISS-ICP：捍卫点对点 ICP – 如果方法正确，注册简单、准确且稳健

伊格纳西奥·维佐·蒂齐亚诺·瓜达尼诺·贝内迪克特·梅尔施·路易斯·威斯曼·詹斯·贝利·西里尔·斯塔尼斯

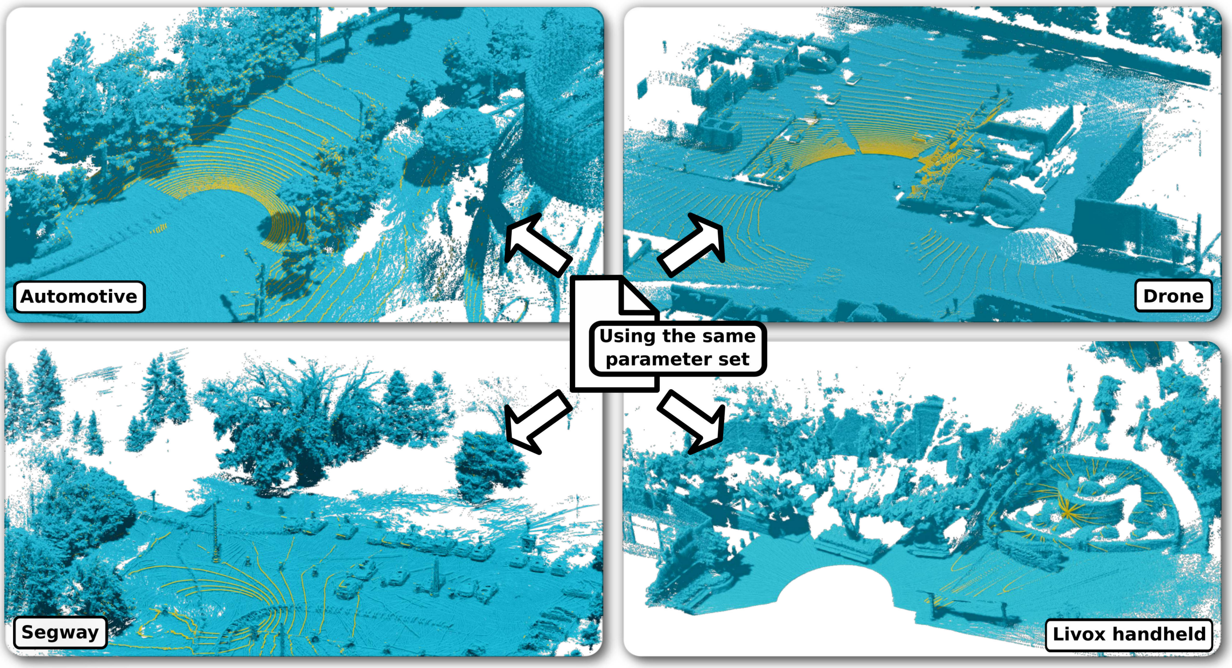

图 1：我们提出的里程计管道在具有相同参数集的不同数据集上生成的点云图（蓝色）。我们用黄色描绘了最新的扫描结果。使用具有不同点密度、不同方向和不同拍摄模式的不同传感器来记录扫描。汽车示例源自 MulRan 数据集 [15]。 Voxgraph 数据集 [23] 的无人机和 NCLT 数据集 [5] 中使用的赛格威机器人显示出高加速度运动曲线。手持式 Livox LiDAR [17] 与常用的旋转机械 LiDAR 具有完全不同的拍摄模式。

Abstract- 机器人平台的稳健且准确的姿态估计，即所谓的基于传感器的里程计，是许多机器人应用的重要组成部分。虽然许多传感器里程计系统通过增加自我运动估计过程的复杂性而取得了进步，但我们却朝着相反的方向前进。通过去除大部分部件并专注于核心元件，我们获得了一个令人惊讶的有效系统，该系统易于实现，并且可以使用不同的激光雷达传感器在各种环境条件下运行。我们的里程计估计方法依赖于点对点 ICP 与对应匹配的自适应阈值、鲁棒内核、简单但广泛适用的运动补偿方法以及点云子采样策略相结合。这产生了一个只有几个参数的系统，在大多数情况下甚至不需要调整到特定的激光雷达传感器。我们的系统在各种操作条件下使用具有相同参数的不同平台，其性能与最先进的方法相当：汽车平台、基于无人机的操作、赛格威等车辆或手持式激光雷达。我们不需要集成 IMU 数据，仅依赖从各种 3D LiDAR 传感器获得的 3D 点云，从而实现广泛的不同应用和操作条件。我们的开源系统的运行速度比所有呈现的数据集中的传感器帧速率都要快，并且专为现实场景而设计。

Index Terms——测绘；本土化; SLAM

一、简介

O DOMETRY 估计是任何需要在未知环境中自主导航的移动机器人的重要构建模块。在 LiDAR 传感领域，当前的里程计管道通常使用某种形式的迭代最近点 (ICP) 来增量估计姿态 [10]、[26]、[31]、[35]。尽管 LiDAR 里程计在过去三十年中一直是一个活跃的研究领域，但当前系统的设计通常与有关机器人运动 [10] 和环境结构 [28] 的假设相结合，以

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achieve accurate and robust alignment results. To the best of our knowledge, no existing 3D LiDAR odometry approach is free of parameter tuning and works out of the box in different scenarios, using arbitrary LiDAR sensors, supporting different motion profiles, and consequently types of robots, such as ground and aerial robots.

This paper returns to the roots: classical point-to-point ICP, introduced 30 years ago by Besl and McKay [3]. We aim to tackle the inherent problems of sequentially operating LiDAR odometry systems that prohibit current approaches from generalizing to different environments, sensor resolutions, and motion profiles using a single configuration. We present simple yet effective reasoning about the robot kinematics and the sequential way LiDAR data is recorded on a mobile platform, as well as an effective downsampled point cloud representation that allows us to minimize the need for parameter tuning.

Our system challenges even extensively hand-tuned and optimized existing simultaneous localization and mapping (SLAM) systems. Our design uses neither sophisticated feature extraction techniques, learning methods, nor loop closures. The same parameter set works in various challenging scenarios such as highway drives of robot cars with many dynamic objects, drone flights, handheld devices, segways, and more. Thus, we take a step back from mainstream research in LiDAR odometry estimation and focus on reducing the components to their essentials. This makes our system perform extraordinarily well in various real-world scenarios, see Fig. 1.

The main contribution of this paper is a simple yet highly effective approach for building LiDAR odometry systems that can accurately compute a robot's pose online while navigating through an environment. We identify the core components and properly evaluate the impact of different modules on such systems. We show that with the proper use of ICP that builds on basic reasoning about the system's physics and the sensor data's nature, we obtain competitive odometry. Besides motion prediction, spatial scan downsampling, and a robust kernel, we introduce an adaptive threshold approach for ICP in the context of robot motion estimation that makes our approach effective and, at the same time, generalizes easily.

We make three key claims: Our "keep it small and simple" approach exploiting point-to-point ICP is (i) on par with state-of-the-art odometry systems, (ii) can accurately compute the robot's odometry in a large variety of environments and motion profiles with the same system configuration, and (iii) provides an effective solution to motion distortion without relying on IMUs or wheel odometers. In sum, "good old point-to-point ICP" is a surprisingly powerful tool, and there is little need to move to more sophisticated approaches if the basic components are done well.

We provide an open-source implementation at: https://github.com/PRBonn/kiss-icp that precisely follows the description of this paper.

II. Related Work

Point cloud registration has been an active area of research for the last three decades [3], [9] and is still relevant nowadays. The ICP algorithm can solve the problem of finding a transformation that brings two different point clouds into a common reference frame, and it is a special case of the absolute orientation problem in photogrammetry. ICP typically consists of two parts. The first one is to find correspondences between the point clouds. The second one computes the transformation that minimizes an objective function defined on the correspondences from the first step. One repeats this process until a convergence criterion is met. Most ICP variants [1], [10], [11], [26], [21], [35] utilize a maximum distance threshold in the data association module plus a robust kernel [6] and a maximum number of iterations. In contrast, we propose a threshold estimation method that adapts to changing scenarios by reasoning about the system kinematics and the nature of the data in combination with a robust kernel. We avoid controlling the number of iterations of the ICP to achieve better generalization.

ICP can be used to obtain an odometry estimation from streaming data from a sensor such as RGB-D cameras [19] or LiDARs [10]. In this work, we focus on the problem of LiDAR odometry estimation, although the ideas presented can be easily extended to other range-sensing technologies.

Nearly all modern SLAM systems build on top of odometry algorithms. Zhang et al. [35] proposed lidar odometry and mapping (LOAM) that computes the robot's odometry by registering planar and edge features to a sparse feature map. LOAM inspired numerous other works [27], [33], such as Lego-LOAM [28], which adds ground constraints to improve accuracy, and recently F-LOAM [33], which revised the original method with a more efficient optimization technique enabling faster operation. However, these methods rely on hand-tuned feature extraction, which typically requires tedious parameter tuning that depends on sensor resolution, environment structure, etc. In contrast, we only rely on point coordinates removing this data-dependent parameter adaptation.

Behley and Stachniss [1] propose the surfel-based method SuMa to achieve LiDAR odometry estimation and mapping. It has also been extended to account for semantics [8] and explicitly handle dynamic objects [7]. In contrast to the surfel-based mapping, Deschaud [11] introduced IMLS-SLAM [11] selecting an implicit moving least square surface [16] as map representation. Along these lines, Vizzo et al. [31] exploited a triangular mesh as the internal map representation. All the above approaches rely on a point-to-plane [24] metric to register consecutive scans. This requires normal estimation, which introduces additional data-dependent parameters. Furthermore, noisy 3D information can impact the normal computation and subsequently the registration in a negative way. We will show that by minimizing a simpler point-to-point metric, we obtain on-par or better odometry performance. Moreover, this design choice enables us to represent the internal map as a voxelized, downsampled point cloud, simplifying the implementation.

Recently, several new approaches [10], [21], [27] have been proposed to solve the odometry estimation problem. Most of these works focus on the runtime operation of the system as well as on the accuracy. Pan et al. [21] propose a multi-metric system (MULLS) that obtains good results in many challenging scenarios at the cost of tuning many parameters for each run. Dellenbach et al. [10] introduced a novel approach, called continuous time ICP (CT-ICP), which incorporates the

获得准确、稳健的对准结果。据我们所知，现有的 3D LiDAR 里程计方法都需要进行参数调整，并且可以在不同场景下开箱即用，使用任意 LiDAR 传感器，支持不同的运动轨迹，从而支持不同类型的机器人，例如地面和空中机器人。

本文回归根源：经典的点对点 ICP，由 Besl 和 McKay 于 30 年前提出[3]。我们的目标是解决顺序操作激光雷达里程计系统的固有问题，这些问题阻止当前方法使用单一配置推广到不同的环境、传感器分辨率和运动轮廓。我们提出了关于机器人运动学和在移动平台上记录 LiDAR 数据的顺序方式的简单而有效的推理，以及有效的下采样点云表示，使我们能够最大限度地减少参数调整的需要。

我们的系统甚至挑战了广泛的手动调整和优化的现有同步定位和地图 (SLAM) 系统。我们的设计既不使用复杂的特征提取技术、学习方法，也不使用闭环。相同的参数集适用于各种具有挑战性的场景，例如具有许多动态物体的机器人汽车在高速公路上行驶、无人机飞行、手持设备、赛格威等。因此，我们从 LiDAR 里程计估计的主流研究中退一步，专注于将组件减少到其本质。这使得我们的系统在各种现实场景中都表现得非常出色，见图 1。

本文的主要贡献是一种简单而高效的方法，用于构建 LiDAR 测距系统，该系统可以在环境中导航时准确地在线计算机器人的姿态。我们识别核心组件并正确评估不同模块对此类系统的影响。我们证明，通过正确使用基于系统物理和传感器数据性质的基本推理的 ICP，我们可以获得有竞争力的里程计。除了运动预测、空间扫描下采样和鲁棒内核之外，我们还在机器人运动估计的背景下引入了 ICP 的自适应阈值方法，这使得我们的方法有效，同时易于推广。

我们提出三个关键主张：我们利用点对点 ICP 的“保持小而简单”的方法 (i) 与最先进的里程计系统相当，(ii) 可以在多种环境和具有相同系统配置的运动曲线中准确计算机器人的里程计，以及 (iii) 提供运动失真的有效解决方案，而无需依赖 IMU 或车轮里程计。总之，“老式的点对点 ICP”是一个令人惊讶的强大工具，如果基本组件做得很好，则几乎不需要转向更复杂的方法。

我们在以下位置提供了一个开源实现：https://github.com/PRBonn/kiss-icp，它完全遵循本文的描述。

二.相关工作

过去三十年来，点云配准一直是一个活跃的研究领域[3]、[9]，并且至今仍然具有相关性。 ICP算法可以解决寻找将两个不同点云转化为一个公共点云的转换问题。参考系，它是摄影测量中绝对方向问题的一个特例。 ICP 通常由两部分组成。第一个是找到点云之间的对应关系。第二个计算最小化在第一步对应关系上定义的目标函数的变换。重复这一过程直到满足收敛标准。大多数 ICP 变体 [1]、[10]、[11]、[26]、[21]、[35] 利用数据关联模块中的最大距离阈值以及鲁棒内核 [6] 和最大迭代次数。相比之下，我们提出了一种阈值估计方法，该方法通过推理系统运动学和数据的性质并结合鲁棒的内核来适应不断变化的场景。我们避免控制 ICP 的迭代次数以实现更好的泛化。

ICP 可用于从来自 RGB-D 相机 [19] 或 LiDAR [10] 等传感器的流数据获取里程计估计。在这项工作中，我们重点关注激光雷达里程计估计的问题，尽管所提出的想法可以很容易地扩展到其他距离传感技术。

几乎所有现代 SLAM 系统都建立在里程计算法之上。张等人。 [35]提出了激光雷达里程计和映射（LOAM），通过将平面和边缘特征注册到稀疏特征图来计算机器人的里程计。 LOAM 启发了许多其他工作 [27]、[33]，例如 Lego-LOAM [28]，它添加了地面约束以提高精度，而最近的 F-LOAM [33]，它使用更有效的优化技术修改了原始方法，从而实现更快的操作。然而，这些方法依赖于手动调整的特征提取，这通常需要依赖于传感器分辨率、环境结构等的繁琐的参数调整。相反，我们只依赖于点坐标来消除这种依赖于数据的参数自适应。

Behley和Stachniss[1]提出了基于面元的方法SuMa来实现LiDAR里程计估计和建图。它还经过扩展以解释语义 [8] 并显式处理动态对象 [7]。与基于面元的映射相比，Deschaud [11] 引入了 IMLS-SLAM [11]，选择隐式移动最小二乘表面 [16] 作为地图表示。沿着这些思路，维佐等人。 [31] 利用三角形网格作为内部地图表示。所有上述方法都依赖于点到平面[24]度量来注册连续扫描。这需要正常估计，这会引入额外的数据相关参数。此外，嘈杂的 3D 信息可能会对正常计算以及随后的配准产生负面影响。我们将证明，通过最小化更简单的点对点度量，我们可以获得同等或更好的里程计性能。此外，这种设计选择使我们能够将内部地图表示为体素化、下采样的点云，从而简化了实现。

最近，提出了几种新方法[10]、[21]、[27]来解决里程估计问题。这些工作大多数都关注系统的运行时操作以及准确性。潘等人。 [21]提出了一种多度量系统（MULLS），该系统在许多具有挑战性的场景中获得了良好的结果，但代价是每次运行都要调整许多参数。德伦巴赫等人。 [10] 介绍了一种称为连续时间 ICP (CT-ICP) 的新颖方法，该方法结合了

motion un-distortion into the registration showing great results but adding more complexity. Additionally, the robots' motion profile must be known a priori, as, for example, a car will have a different profile than a segway platform. We challenge the need for sophisticated optimization techniques to cope with motion distortion requiring only the constant velocity model. Furthermore, our system only relies on a few parameters, and we do not need to know the motion profile in advance.

Many state-of-the-art odometry systems [1], [10], [21], [27] also rely on pose graph optimization to achieve a better alignment. In contrast, we do not exploit such techniques and state that pose graph optimization is orthogonal to the presented approach and can be easily integrated. In sum, we step back from the common mainstream work on LiDAR odometry and propose a system that solely relies on a point-to-point metric and does not employ pose graph optimization [1], [10], [21], [27]. Our system can run on different types of mobile robots, drones, handheld devices, and segways, without the need to fine-tune the system to a specific application.

III. KISS-ICP – Keep It Small and Simple

This work aims to incrementally compute the trajectory of a moving LiDAR sensor by sequentially registering the point clouds recorded by the scanner. We reduced the components to a minimal set needed to build an effective, accurate, robust, and still reasonably simple LiDAR odometry system.

For each 3D scan in form of a local, egocentric point cloud $\mathcal{P} = \{\boldsymbol{p}_i \mid \boldsymbol{p}_i \in \mathbb{R}^3\}$, we perform the following four steps to obtain a global pose estimate $\mathsf{T}_t \in SE(3)$ at time t. First, we apply sensor motion prediction and motion compensation, often called deskewing, to undo the distortions of the 3D data caused by the sensor's motion during scanning. Second, we subsample the current scan. Third, we estimate correspondences between the input point cloud and a reference point cloud, which we call the local map. We use an adaptive thresholding scheme for correspondence estimation, restricting possible data associations and filtering out potential outliers. Fourth, we register the input point cloud to the local map using a robust point-to-point ICP algorithm. Finally, we update the local map with a downsampled version of the registered scan. Below, we describe these components in detail.

A. Step 1: Motion Prediction and Scan Deskewing

We advocate for rethinking the point cloud registration in the context of mobile robots, which continuously record data. One should not think of it as registering arbitrary pairs of 3D point clouds. Instead, one should phrase it as *estimating how much the robot's actual motion deviates from its expected motion by registering consecutive scans*.

Different approaches can be used to compute the robot's expected motion before considering the LiDAR data. The three most popular choices are the constant velocity model, wheel odometry obtained through encoders, and IMU-based motion estimation. The constant velocity [29] model assumes that a robot moves with the same translational and rotational velocity as in the previous time step. It requires no additional sensors (no wheel encoder, no IMU) and thus is the most widely applicable option.

Our approach uses the constant velocity model for two reasons: first, it is generally applicable, requires no additional sensors, and avoids the need for time synchronization between sensors. Second, as we will show in our experimental evaluation, it works well enough to provide a solid initial guess when searching for data associations and deskewing 3D scans. This follows from the fact that robotic LiDAR sensors commonly record and stream point clouds at 10 Hz to 20 Hz, i.e., every 0.05 s to 0.1 s. In most cases, the acceleration or deceleration, i.e., the deviations from the constant velocity model that occurs within such short time intervals, are fairly small. If the robot accelerates or decelerates, the constant velocity estimation of the robot's pose will be slightly off, and therefore, we need to correct this estimate through registration. These accelerations determine the possible displacements of the (static) 3D points.

The constant velocity model approximates the translational and angular velocities, denoted as $\boldsymbol{v}_t$ and $\boldsymbol{\omega}_t$ at time t respectively, by using the previous pose estimates $\mathsf{T}_{t-1} = (\boldsymbol{R}_{t-1}, \boldsymbol{t}_{t-1})$ and $\mathsf{T}_{t-2} = (\boldsymbol{R}_{t-2}, \boldsymbol{t}_{t-2})$, represented by a rotation matrix $\boldsymbol{R}_t \in SO(3)$ and a translation vector $\boldsymbol{t}_t \in \mathbb{R}^3$ for the time step t. We first compute the relative pose $\mathsf{T}_{\text{pred},t}$ that we will use as motion prediction as:

$$\mathsf{T}_{\text{pred},t} = \begin{bmatrix} \boldsymbol{R}_{t-2}^\top \boldsymbol{R}_{t-1} & \boldsymbol{R}_{t-2}^\top (\boldsymbol{t}_{t-1} - \boldsymbol{t}_{t-2}) \\ \boldsymbol{0} & 1 \end{bmatrix}, \tag{1}$$

then derive the corresponding velocities as:

$$\boldsymbol{v}_t = \frac{\boldsymbol{R}_{t-2}^\top (\boldsymbol{t}_{t-1} - \boldsymbol{t}_{t-2})}{\Delta t}, \tag{2}$$

$$\boldsymbol{\omega}_t = \frac{\mathrm{Log}(\boldsymbol{R}_{t-2}^\top \boldsymbol{R}_{t-1})}{\Delta t}, \tag{3}$$

where Δt is the acquisition time of one LiDAR sweep, typically 0.05 s or 0.1 s, and Log: $SO(3) \to \mathbb{R}^3$ extracts the axis-angle representation.

Note that also wheel odometry or an IMU-based motion prediction approach can be used instead to compute $\boldsymbol{v}_t$ and $\boldsymbol{\omega}_t$ for each time step. This will not change the remainder of our approach. For example, if one has good wheel odometry available, this can also be used. However, we use constant velocity as a generally applicable approach.

Within the acquisition time Δt of one LiDAR sweep, multiple 3D points are measured by the scanner. The relative timestamp $s_i \in [0, \Delta t]$ for each point $\boldsymbol{p}_i \in \mathcal{P}$ describes the recording time relative to the scan's first measurement. This relative timestamp allows us to compute the motion compensation resulting in a deskewed point $\boldsymbol{p}_i^* \in \mathcal{P}^*$ of the corrected scan $\mathcal{P}^*$ reading by

$$\boldsymbol{p}_i^* = \mathrm{Exp}(s_i \boldsymbol{\omega}_t)\boldsymbol{p}_i + s_i \boldsymbol{v}_t, \tag{4}$$

where Exp: $\mathbb{R}^3 \to SO(3)$ computes a rotation matrix from an axis-angle representation. Note that $\mathrm{Exp}(s_i \boldsymbol{\omega}_t)$ is equivalent to performing SLERP in the axis-angle domain.

This form of scan deskewing, especially with the constant velocity model, is easy to implement, generally applicable, and does not require additional sensors, high-precision time synchronization between sensors, or IMU biases to be estimated.

运动不失真注册显示出很好的结果，但增加了更多的复杂性。此外，机器人的运动轮廓必须先验已知，例如，汽车将具有与赛格威平台不同的轮廓。我们挑战需要复杂的优化技术来应对仅需要恒速模型的运动失真。此外，我们的系统仅依赖于几个参数，并且我们不需要提前知道运动曲线。

许多最先进的里程计系统 [1]、[10]、[21]、[27] 也依赖位姿图优化来实现更好的对齐。相反，我们没有利用此类技术，并声明位姿图优化与所提出的方法正交并且可以轻松集成。总之，我们从 LiDAR 里程计的常见主流工作中退一步，提出了一种仅依赖于点对点度量且不采用位姿图优化的系统[1]、[10]、[21]、[27]。我们的系统可以在不同类型的移动机器人、无人机、手持设备和赛格威上运行，无需针对特定应用程序对系统进行微调。

三. KISS-ICP – 保持小而简单

这项工作的目的是通过顺序注册扫描仪记录的点云来增量计算移动激光雷达传感器的轨迹。我们将构建有效、准确、稳健且仍然相当简单的 LiDAR 里程计系统所需的组件减少到最少。

对于本地、以自我为中心的点云 $\mathcal{P} = \{\boldsymbol{p}_i \mid \boldsymbol{p}_i \in \mathbb{R}^3\}$ 形式的每次 3D 扫描，我们执行以下四个步骤以获得时间 t 的全局姿态估计 $\mathrm{T}_t \in SE(3)$。首先，我们应用传感器运动预测和运动补偿（通常称为纠偏）来消除扫描过程中传感器运动引起的 3D 数据失真。其次，我们对当前扫描进行二次采样。第三，我们估计输入点云和参考点云（我们称之为局部地图）之间的对应关系。我们使用自适应阈值方案进行对应估计，限制可能的数据关联并过滤掉潜在的异常值。第四，我们使用稳健的点对点 ICP 算法将输入点云注册到本地地图。最后，我们使用注册扫描的下采样版本更新本地地图。下面，我们详细描述这些组件。

A. Step 1: Motion Prediction and Scan Deskewing

我们主张在不断记录数据的移动机器人的背景下重新考虑点云注册。人们不应将其视为注册任意对的 3D 点云。相反，应该将其表述为 *estimating how much the robot's actual motion deviates from its expected motion by registering consecutive scans*。

在考虑激光雷达数据之前，可以使用不同的方法来计算机器人的预期运动。三种最流行的选择是等速模型、通过编码器获得的车轮里程计以及基于 IMU 的运动估计。恒定速度[29]模型假设机器人以与前一个时间步相同的平移和旋转速度移动。不需要额外的传感器（无车轮编码器，无 IMU），因此是应用最广泛的选项。

我们的方法使用等速模型有两个原因：首先，它具有普遍适用性，不需要额外的传感器，并且避免了传感器之间时间同步的需要。其次，正如我们将在实验评估中展示的那样，它运行良好，足以在搜索数据关联和校正 3D 扫描时提供可靠的初始猜测。这是因为机器人 LiDAR 传感器通常以 10 Hz 至 20 Hz（即每 0.05 秒至 0.1 秒）记录和传输点云。在大多数情况下，加速度或减速度，即在如此短的时间间隔内发生的与等速模型的偏差，是相当小的。如果机器人加速或减速，机器人位姿的等速估计会略有偏差，因此，我们需要通过配准来修正这个估计。这些加速度决定了（静态）3D 点可能的位移。

等速模型通过使用先前的姿态估计 $\mathrm{T}_{t-1} = (\boldsymbol{R}_{t-1}, \boldsymbol{t}_{t-1})$ 和 $\mathrm{T}_{t-2} = (\boldsymbol{R}_{t-2}, \boldsymbol{t}_{t-2})$（由时间步长 t 的旋转矩阵 $\boldsymbol{R}_t \in SO(3)$ 和平移向量 $\boldsymbol{t}_t \in \mathbb{R}^3$ 表示）来近似平移速度和角速度，在时间 t 分别表示为 $\boldsymbol{v}_t$ 和 $\boldsymbol{\omega}_t$。我们首先计算将用作运动预测的相对姿势 $\mathrm{T}_{\mathrm{pred},t}$，如下所示：

$$\mathrm{T}_{\mathrm{pred},t} = \begin{bmatrix} \boldsymbol{R}_{t-2}^\top \boldsymbol{R}_{t-1} & \boldsymbol{R}_{t-2}^\top (\boldsymbol{t}_{t-1} - \boldsymbol{t}_{t-2}) \\ \boldsymbol{0} & 1 \end{bmatrix}, \tag{1}$$

然后推导相应的速度为：

$$\boldsymbol{v}_t = \frac{\boldsymbol{R}_{t-2}^\top (\boldsymbol{t}_{t-1} - \boldsymbol{t}_{t-2})}{\Delta t}, \tag{2}$$

$$\boldsymbol{\omega}_t = \frac{\mathrm{Log}(\boldsymbol{R}_{t-2}^\top \boldsymbol{R}_{t-1})}{\Delta t}, \tag{3}$$

其中 Δt 是一次 LiDAR 扫描的采集时间，通常为 0.05 s 或 0.1 s，Log: $SO(3) \to \mathbb{R}^3$ 提取轴角表示。

请注意，也可以使用车轮里程计或基于 IMU 的运动预测方法来计算每个时间步的 $\boldsymbol{v}_t$ 和 $\boldsymbol{\omega}_t$。这不会改变我们方法的其余部分。例如，如果有良好的车轮里程计可用，也可以使用它。然而，我们使用恒定速度作为普遍适用的方法。

在一次 LiDAR 扫描的采集时间 Δt 内，扫描仪测量多个 3D 点。每个点 $\boldsymbol{p}_i \in \mathcal{P}$ 的相对时间戳 $s_i \in [0, \Delta t]$ 描述相对于扫描第一次测量的记录时间。这个相对时间戳允许我们计算运动补偿，从而产生校正扫描 $\mathcal{P}^*$ 读数的偏斜校正点 $\boldsymbol{p}_i^* \in \mathcal{P}^*$

$$\boldsymbol{p}_i^* = \mathrm{Exp}(s_i \boldsymbol{\omega}_t) \boldsymbol{p}_i + s_i \boldsymbol{v}_t, \tag{4}$$

其中 Exp: $\mathbb{R}^3 \to SO(3)$ 根据轴角度表示计算旋转矩阵。请注意，$\mathrm{Exp}(s_i \boldsymbol{\omega}_t)$ 相当于在轴角度域中执行 SLERP。

这种形式的扫描偏移校正，尤其是恒速模型，易于实现，普遍适用，并且不需要额外的传感器、传感器之间的高精度时间同步或估计IMU偏差。

As we show in Sec. IV, this approach often performs even better than more complex compensation systems [10], at least as long the motion between the start and end of the sweep is small as it is for most robotics applications.

B. Step 2: Point Cloud Subsampling

Identifying a set of keypoints in the point cloud is a common approach for scan registration [14], [24], [35]. It is typically done to achieve faster convergence and/or higher robustness in the data association. However, complex filtering of the point cloud usually comes with an extra layer of complexity and parameters that often need to be tuned.

Rather than extracting 3D keypoints, which often requires environment-dependent parameter tuning, we propose to compute only a spatially downsampled version $\hat{\mathcal{P}}^*$ of the deskewed scan $\mathcal{P}^*$. Downsampling is done using a voxel grid. As we will explain in Sec. III-C below in more detail, we use a voxel grid as our local map, where each voxel call has a size of $v \times v \times v$ and each cell only store a certain number of points. Every time we process an incoming scan, we first downsample the point cloud of the scan to an intermediate point cloud $\mathcal{P}^*_{\text{merge}}$, which is later used to update the map when the relative motion of the robot has been determined with ICP. To obtain the points in $\mathcal{P}^*_{\text{merge}}$, we use voxel size $\alpha\, v$ with $\alpha \in (0.0, 1.0]$ and keep only a single point per voxel.

For the ICP registration, an even lower resolution scan is beneficiary. Thus, we compute a further reduced point cloud $\hat{\mathcal{P}}^*$ by downsampling $\mathcal{P}^*_{\text{merge}}$ again using a voxel size of $\beta\, v$ with $\beta \in [1.0, 2.0]$ keeping only a single point per voxel. This further reduces the number of points processed during the registration and allows for a fast and highly effective alignment. The idea of this “double downsampling” stems from CT-ICP [10], the so far best performing open-source LiDAR odometry system on KITTI.

Most voxelization approaches, however, select the center of each occupied voxel to downsample the point cloud [25], [36]. Instead, we found it advantageous to maintain the original point coordinates, select only one point per voxel for a single scan, and keep its coordinates to avoid discretization errors. This means the reduced cloud is a subset of the deskewed one, i.e., $\hat{\mathcal{P}}^* \subseteq \mathcal{P}^*$. In our implementation, we keep only the first point that was inserted into the voxel.

C. Step 3: Local Map and Correspondence Estimation

In line with prior work [1], [10], [19], [35], we register the deskewed and subsampled scan $\hat{\mathcal{P}}^*$ to the point cloud built so far, i.e., a local map, to compute an incremental pose estimate $\Delta \mathsf{T}_{\text{icp}}$. We use frame-to-map registration as it proves more reliable and robust than the frame-to-frame alignment [1], [19]. To do that effectively, we must define a data structure representing the previously registered scans.

Modern approaches have used very different types of representations for this local map. Popular approaches are voxel grids [35], triangle meshes [31], surfel representations [1], or implicit representations [11]. As mentioned in Sec. III-B, we utilize a voxel grid to store a subset of 3D points. We use a grid with a voxel size of $v \times v \times v$ and store up to $N_{\max}$ points per voxel. After registration, we update the voxel grid by adding the points $\{\mathsf{T}_t\, \boldsymbol{p} \,|\, \boldsymbol{p} \in \mathcal{P}^*_{\text{merge}}\}$ from the new scan using the global pose estimate T_t. Voxels that already contain $N_{\max}$ points are not updated. Additionally, given the current pose estimate, we remove voxels outside the maximum range $r_{\max}$. Thus, the size of the map will stay bounded.

Instead of a 3D array, we use a hash table to store the voxels, allowing a memory-efficient representation and fast nearest neighbor search [10], [20]. However, the used data structure can be easily replaced with VDBs [18], [32], Octrees [30], [34], or KD-Trees [2].

D. Adaptive Threshold for Data Association

ICP typically performs a nearest neighbor data association to find corresponding points between two point clouds [3]. When searching for associations, it is common to impose a maximum distance between corresponding points, often using a value of 1 m or 2 m [1], [31], [35]. This maximum distance threshold can be seen as an outlier rejection scheme, as all correspondences with a distance larger than this threshold are considered outliers and are ignored.

The required value for this threshold τ depends on the expected initial pose error, the number and type of dynamic objects in the scene, and, to some degree, the sensor noise. It is typically selected heuristically. Based on the considerations about the constant velocity motion prediction in Sec. III-A, we can, however, estimate a likely limit from data by analyzing how much the odometry may deviate from the motion prediction over time. This deviation $\Delta\mathsf{T}$ in the pose corresponds exactly to the local ICP correction to be applied to the predicted pose (but it is not known beforehand). Intuitively, we can observe the robot’s acceleration in the magnitude of $\Delta\mathsf{T}$. If the robot is not accelerating, then $\Delta\mathsf{T}$ will have a small magnitude, often around zero, meaning that the constant velocity assumption holds and no correction has to be done by ICP.

We integrate this information into our data association search by exploiting the so-far successful ICP executions. We can estimate the possible point displacement between corresponding points in successive scans in the presence of a potential acceleration expressed through $\Delta\mathsf{T}$ as:

$$\delta(\Delta\mathsf{T}) = \delta_{\text{rot}}(\Delta \boldsymbol{R}) + \delta_{\text{trans}}(\Delta \boldsymbol{t}), \tag{5}$$

where $\Delta \boldsymbol{R} \in SO(3)$ and $\Delta \boldsymbol{t} \in \mathbb{R}^3$ refer to the rotational and translational component of the deviation, given by

$$\delta_{\text{rot}}(\Delta \boldsymbol{R}) = 2\; r_{\max} \sin\left(\frac{1}{2} \underbrace{\arccos\left(\frac{\operatorname{tr}(\Delta \boldsymbol{R}) - 1}{2}\right)}_{\theta}\right) \tag{6}$$

$$\delta_{\text{trans}}(\Delta \boldsymbol{t}) = \|\Delta \boldsymbol{t}\|_2. \tag{7}$$

The term $\delta_{\text{rot}}(\Delta \boldsymbol{R})$ represents the displacement that occurs for a range reading with maximum range $r_{\max}$ subject to the rotation $\Delta \boldsymbol{R}$, see also Fig. 2. Note that Eq. (5) constitutes an upper bound for the point displacement as

$$\|\Delta \boldsymbol{R}\, \boldsymbol{p} + \Delta \boldsymbol{t} - \boldsymbol{p}\|_2 \leq \delta_{\text{rot}}(\Delta \boldsymbol{R}) + \delta_{\text{trans}}(\Delta \boldsymbol{t}), \tag{8}$$

which follows from the triangle inequality.

正如我们在第二节中所示。 IV，这种方法通常比更复杂的补偿系统表现更好[10]，至少只要扫描开始和结束之间的运动很小，就像大多数机器人应用一样。

B. Step 2: Point Cloud Subsampling

识别点云中的一组关键点是扫描配准的常用方法[14]、[24]、[35]。这样做通常是为了在数据关联中实现更快的收敛和/或更高的鲁棒性。然而，点云的复杂过滤通常会带来额外的复杂性和经常需要调整的参数。

我们建议只计算去歪斜扫描 $\mathcal{P}^*$ 的空间下采样版本$\hat{\mathcal{P}}^*$，而不是提取通常需要依赖于环境的参数调整的 3D 关键点。下采样是使用体素网格完成的。正如我们将在第 2 节中解释的那样。下面的 III-C 更详细地介绍了，我们使用体素网格作为局部地图，其中每个体素调用的大小为 $v \times v \times v$，并且每个单元仅存储一定数量的点。每次我们处理传入的扫描时，我们首先将扫描的点云下采样为中间点云 $\mathcal{P}^*_{\text{merge}}$，稍后在通过 ICP 确定机器人的相对运动时用于更新地图。为了获得 $\mathcal{P}^*_{\text{merge}}$ 中的点，我们使用体素大小 $\alpha\, v$ 和 $\alpha \in (0.0, 1.0]$ 并为每个体素仅保留一个点。

对于 ICP 注册，分辨率更低的扫描也是受益的。因此，我们通过使用体素大小 $\beta\, v$ 再次下采样 $\mathcal{P}^*_{\text{merge}}$ 来计算进一步缩小的点云$\hat{\mathcal{P}}^*$，其中 $\beta \in [1.0, 2.0]$ 每个体素仅保留一个点。这进一步减少了配准过程中处理的点的数量，并允许快速高效的对准。这种“双下采样”的想法源于 CT-ICP [10]，它是 KITTI 上迄今为止性能最好的开源 LiDAR 测距系统。

然而，大多数体素化方法选择每个占用体素的中心来对点云进行下采样[25]、[36]。相反，我们发现保持原始点坐标是有利的，对于单次扫描每个体素仅选择一个点，并保留其坐标以避免离散化错误。这意味着减少的云是去歪斜云的子集，即$\hat{\mathcal{P}}^* \subseteq \mathcal{P}^*$。在我们的实现中，我们仅保留插入到体素中的第一个点。

C. Step 3: Local Map and Correspondence Estimation

根据之前的工作[1]、[10]、[19]、[35]，我们将去歪斜和子采样扫描$\hat{\mathcal{P}}^*$注册到迄今为止构建的点云（即局部地图），以计算增量姿态估计$\Delta \mathrm{T}_{\text{icp}}$。我们使用帧到地图配准，因为它比帧到帧对齐更可靠和鲁棒[1]，[19]。为了有效地做到这一点，我们必须定义一个表示先前注册的扫描的数据结构。

现代方法对这张本地地图使用了非常不同类型的表示形式。流行的方法是体素网格[35]、三角形网格[31]、面元表示[1]或隐式表示[11]。正如第 2 节中提到的。 III-B，我们利用体素网格来存储 3D 点的子集。我们使用体素大小为 $v \times v \times v$ 的网格并存储最多 N_{max} 每个体素的点。配准后，我们通过使用全局姿态估计 T_t 添加新扫描中的点 $\{\mathrm{T}_t\, \boldsymbol{p} \mid \boldsymbol{p} \in \mathcal{P}^*_{\text{merge}}\}$ 来更新体素网格。已包含 N_{max} 点的体素不会更新。此外，考虑到当前的姿态估计，我们删除了最大范围 r_{max} 之外的体素。因此，地图的大小将保持有限。

我们使用哈希表来存储体素，而不是 3D 数组，从而实现内存高效表示和快速最近邻搜索 [10]、[20]。然而，所使用的数据结构可以很容易地替换为 VDB [18]、[32]、八叉树 [30]、[34] 或 KD 树 [2]。

D. Adaptive Threshold for Data Association

ICP 通常执行最近邻数据关联来查找两个点云之间的对应点 [3]。在搜索关联时，通常会在对应点之间施加最大距离，通常使用 1 m 或 2 m 的值 [1]、[31]、[35]。该最大距离阈值可以被视为异常值拒绝方案，因为距离大于该阈值的所有对应都被视为异常值并被忽略。

该阈值 τ 所需的值取决于预期的初始姿态误差、场景中动态对象的数量和类型，以及在某种程度上传感器噪声。它通常是启发式选择的。基于第 2 节中关于等速运动预测的考虑。然而，III-A，我们可以通过分析里程计随着时间的推移可能偏离运动预测的程度来估计数据的可能限制。位姿中的偏差 ΔT 与应用于预测位姿的局部 ICP 校正完全对应（但事先并不知道)）。直观地，我们可以观察机器人的加速度，其幅度为 ΔT。如果机器人没有加速，则 ΔT 将具有较小的幅度，通常在零左右，这意味着恒定速度假设成立，并且无需通过 ICP 进行校正。

我们通过利用迄今为止成功的 ICP 执行，将此信息集成到我们的数据关联搜索中。我们可以估计在存在潜在加速度的情况下连续扫描中对应点之间可能的点位移，通过 ΔT 表示为：

$$\delta(\Delta \mathsf{T}) = \delta_{\text{rot}}(\Delta \boldsymbol{R}) + \delta_{\text{trans}}(\Delta \boldsymbol{t}), \tag{5}$$

其中 $\Delta \boldsymbol{R} \in SO(3)$ 和 $\Delta \boldsymbol{t} \in \mathbb{R}^3$ 指偏差的旋转和平移分量，由下式给出

$$\delta_{\text{rot}}(\Delta \boldsymbol{R}) = 2\ r_{\text{max}} \sin\left(\underbrace{\frac{1}{2} \arccos\left(\frac{\operatorname{tr}(\Delta \boldsymbol{R}) - 1}{2}\right)}_{\theta}\right) \tag{6}$$

$$\delta_{\text{trans}}(\Delta \boldsymbol{t}) = \|\Delta \boldsymbol{t}\|_2. \tag{7}$$

项 $\delta_{\text{rot}}(\Delta \boldsymbol{R})$ 表示最大范围 r_{max} 受到旋转 $\Delta \boldsymbol{R}$ 影响的范围读数时发生的位移，另请参见图 2。 (5) 构成点位移的上限为

$$\|\Delta \boldsymbol{R}\, \boldsymbol{p} + \Delta \boldsymbol{t} - \boldsymbol{p}\|_2 \leq \delta_{\text{rot}}(\Delta \boldsymbol{R}) + \delta_{\text{trans}}(\Delta \boldsymbol{t}), \tag{8}$$

由三角不等式得出。

For obtaining δ_{rot} other approaches could be considered, like taking into account the individual ranges for the adaptive threshold computation [4]. In our tests, we did not see any difference in the results but a 3-fold increase of the overall runtime; thus, we use $r_{\max}$ instead of r for computing δ_{rot}.

To compute the threshold τ_t at time t, we consider a Gaussian distribution over δ using the values of Eq. (5) over the trajectory computed so far whenever the deviation was larger than a minimum distance $\delta_{\min}$, i.e., situations where the robot's motion was deviating from the constant velocity model. Its standard deviation is

$$\sigma_t = \sqrt{\frac{1}{|\mathcal{M}_t|} \sum_{i \in \mathcal{M}_t} \delta(\Delta T_i)^2}, \tag{9}$$

where the index set $\mathcal{M}_t$ of deviations up to t is given by

$$\mathcal{M}_t = \{ i \mid i < t \wedge \delta(\Delta T_i) > \delta_{\min} \}. \tag{10}$$

This avoids reducing the value of σ_t too much when the robot is not moving or is moving at constant velocity for a long time. In our experiments, we set this threshold $\delta_{\min}$ to 0.1 m. We then compute the threshold τ_t as the three-sigma bound $\tau_t = 3\,\sigma_t$, which we use in the next section for the data association search.

E. Step 4: Alignment Through Robust Optimization

We base our registration on classic point-to-point ICP [3]. The advantage of this choice is that we do not need to compute data-dependent features such as normals, curvature, or other descriptors, which may depend on the scanner or the environment. Furthermore, with noisy or sparse LiDAR scanners, features such as normals are often not very reliable. Thus, neglecting quantities such as normals in the alignment process is an explicit design decision that allows our system to generalize well to different sensor resolutions.

To obtain the global estimation of the pose T_t of the robot, we start by applying our prediction model $\mathsf{T}_{\text{pred},t}$ to the scan $\hat{\mathcal{P}}^*$ in the local frame. Successively, we transform it into the global coordinate frame using the previous pose estimate T_{t-1}, resulting in the source points

$$\mathcal{S} = \left\{ \boldsymbol{s}_i = \mathsf{T}_{t-1} \mathsf{T}_{\text{pred},t}\, \boldsymbol{p} \mid \boldsymbol{p} \in \hat{\mathcal{P}}^* \right\}. \tag{11}$$

For each iteration j of ICP, we obtain a set of correspondences between the point cloud $\mathcal{S}$ and the local map $\mathcal{Q} = \{ \boldsymbol{q}_i \mid \boldsymbol{q}_i \in \mathbb{R}^3 \}$ through nearest neighbor search over the voxel grid (Sec. III-C) considering only correspondences with a point-to-point distance below τ_t. To compute the current pose correction $\Delta\mathsf{T}_{\text{est},j}$, we perform a robust optimization minimizing the sum of point-to-point residuals

$$\Delta\mathsf{T}_{\text{est},j} = \underset{\mathsf{T}}{\operatorname{argmin}} \sum_{(s,q) \in \mathcal{C}(\tau_t)} \rho(\| \mathsf{T}\boldsymbol{s} - \boldsymbol{q} \|_2), \tag{12}$$

where $\mathcal{C}(\tau_t)$ is the set of nearest neighbor correspondences with a distance smaller than τ_t and ρ is the Geman-McClure robust kernel, i.e., an M-estimator with a strong outlier rejection property, given by

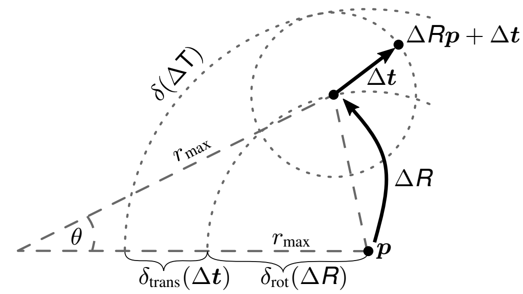

Fig. 2: Exemplary computation of the maximum point displacement $\delta(\Delta\mathsf{T})$ caused by a rotational and translational deviation $(\Delta R, \Delta\boldsymbol{t})$ from the predicted motion.

$$\rho(e) = \frac{e^2/2}{\underbrace{\sigma_t/3}_{\kappa_t} + e^2}, \tag{13}$$

where the scale parameter κ_t of the kernel is adapted online using σ_t. Lastly, we update the points $\boldsymbol{s}_i$, i.e.,

$$\{ \boldsymbol{s}_i \leftarrow \Delta\mathsf{T}_{\text{est},j} \boldsymbol{s}_i \mid \boldsymbol{s}_i \in \mathcal{S} \}, \tag{14}$$

and repeat the process until the convergence criterion is met.

As a result of this process, we obtain the transformation $\mathsf{T}_t = \Delta\mathsf{T}_{\text{icp},t} \mathsf{T}_{t-1} \mathsf{T}_{\text{pred},t}$, where $\Delta\mathsf{T}_{\text{icp},t} = \prod_j \Delta\mathsf{T}_{\text{est},j}$. While we apply the prediction model $\mathsf{T}_{\text{pred},t}$ (i.e., the constant velocity prediction) to the local coordinate frame of the scan, we perform the ICP correction $\Delta\mathsf{T}_{\text{icp},t}$ in the global reference frame of the robot. This is done for efficiency reasons as it allows us to transform the source points $\mathcal{S}$ only once per ICP iteration. With this, the local pose deviation $\Delta\mathsf{T}_t$ at time t used in the Eq. (5) can be expressed as

$$\Delta\mathsf{T}_t = (\mathsf{T}_{t-1}\mathsf{T}_{\text{pred},t})^{-1}\, \Delta\mathsf{T}_{\text{icp},t} \mathsf{T}_{t-1} \mathsf{T}_{\text{pred},t}. \tag{15}$$

A standard termination criterion for the ICP algorithm is to control the number of iterations. Additionally, most approaches also have a further criterion based on the minimum change in the solution. Conversely, we found that controlling the number of iterations does not allow the algorithm to always find a good solution. Thus, we only employ the termination criterion based on the applied correction being smaller than γ, without imposing a maximum number of iterations.

Finally, the ICP correction is applied to the point cloud $\mathcal{P}^*_{\text{merge}}$, and the points are integrated into the local map.

F. Parameters

Our implementation depends on a small set of seven parameters. All are shown in Tab. I. We use the same parameters for all experiments. Most other approaches use a substantially larger set of parameters: MULLS [21] has 107 parameters, SuMa [1] has 49 parameters, and CT-ICP [10] has 30 parameters in their respective configuration files. In contrast, our approach only has two parameters for the correspondence search, four for the map representation and scan subsampling, and one for the ICP termination. Note that the maximum range of a scanner is a value that depends on the specific sensor in use and, as such, we do not consider it a system parameter.

为了获得 δ_{rot}，可以考虑其他方法，例如考虑自适应阈值计算的各个范围[4]。在我们的测试中，我们没有看到结果有任何差异，但整体运行时间增加了 3 倍；因此，我们使用 r_{max} 而不是 r 来计算 δ_{rot}。

为了计算时间 t 处的阈值 τ_t，我们使用等式 1 的值考虑 δ 上的高斯分布。（5）只要偏差大于最小距离δ_{min}，即机器人的运动偏离恒速模型的情况，就计算出迄今为止计算的轨迹。其标准差为

$$\sigma_t = \sqrt{\frac{1}{|\mathcal{M}_t|}\sum_{i\in\mathcal{M}_t}\delta(\Delta T_i)^2}, \tag{9}$$

其中，直到 t 的偏差的索引集 $\mathcal{M}_t$ 由下式给出

$$\mathcal{M}_t = \{i \mid i < t \wedge \delta(\Delta T_i) > \delta_{\text{min}}\}. \tag{10}$$

这样可以避免机器人长时间不动或匀速运动时，将σ_t的值减小太多。在我们的实验中，我们将此阈值 δ_{min} 设置为 0.1 m。然后，我们将阈值 τ_t 计算为三西格玛边界 $\tau_t = 3\ \sigma_t$，我们将在下一节中使用它进行数据关联搜索。

E. Step 4: Alignment Through Robust Optimization

我们的注册基于经典的点对点 ICP [3]。这种选择的优点是我们不需要计算与数据相关的特征，例如法线、曲率或其他描述符，这些特征可能取决于扫描仪或环境。此外，对于嘈杂或稀疏的激光雷达扫描仪，法线等特征通常不太可靠。因此，在对齐过程中忽略诸如法线之类的量是一个明确的设计决策，使我们的系统能够很好地推广到不同的传感器分辨率。

为了获得机器人姿态 T_t 的全局估计，我们首先将预测模型 $\mathrm{T}_{\text{pred},t}$ 应用于局部框架中的扫描$\hat{\mathcal{P}}^*$。接下来，我们使用之前的位姿估计 T_{t-1} 将其转换为全局坐标系，从而得到源点

$$\mathcal{S} = \left\{\boldsymbol{s}_i = \mathsf{T}_{t-1}\mathsf{T}_{\text{pred},t}\,\boldsymbol{p} \mid \boldsymbol{p} \in \hat{\mathcal{P}}^*\right\}. \tag{11}$$

对于 ICP 的每次迭代 j，我们通过体素网格上的最近邻搜索获得点云 $\mathcal{S}$ 和局部地图 $\mathcal{Q} = \{\boldsymbol{q}_i \mid \boldsymbol{q}_i \in \mathbb{R}^3\}$ 之间的一组对应关系（第 III-C 节），仅考虑点到点距离低于 τ_t 的对应关系。为了计算当前姿态校正 $\Delta\mathrm{T}_{\text{est},j}$，我们执行稳健的优化，最小化点对点残差之和

$$\Delta\mathsf{T}_{\text{est},j} = \underset{\mathsf{T}}{\operatorname{argmin}} \sum_{(s,q)\in\mathcal{C}(\tau_t)} \rho(\|\mathsf{T}\boldsymbol{s} - \boldsymbol{q}\|_2), \tag{12}$$

其中 $\mathcal{C}(\tau_t)$ 是距离小于 τ_t 的最近邻对应集，ρ 是 Geman-McClure

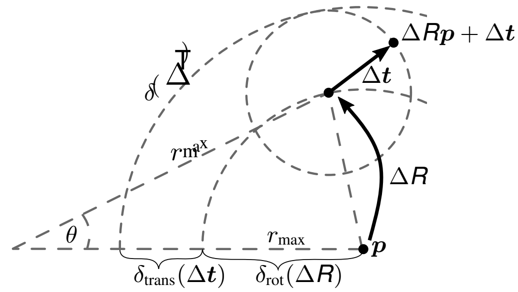

图 2：由预测运动的旋转和平移偏差 $(\Delta R, \Delta\boldsymbol{t})$ 引起的最大点位移 $\delta(\Delta\mathrm{T})$ 的示例性计算。

鲁棒核，即具有强异常值拒绝特性的 M 估计器，由下式给出

$$\rho(e) = \frac{e^2/2}{\underbrace{\sigma_t/3}_{\kappa_t} + e^2}, \tag{13}$$

其中内核的尺度参数 κ_t 使用 σ_t 在线调整。最后，我们更新点 $\boldsymbol{s}_i$，即

$$\{\boldsymbol{s}_i \leftarrow \Delta\mathsf{T}_{\text{est},j}\boldsymbol{s}_i \mid \boldsymbol{s}_i \in \mathcal{S}\}, \tag{14}$$

并重复该过程，直到满足收敛标准

。此过程的结果是，我们获得变换 $\mathrm{T}_t = \Delta\mathrm{T}_{\text{icp},t}\mathrm{T}_{t-1}\mathrm{T}_{\text{pred},t}$，其中 $\Delta\mathrm{T}_{\text{icp},t} = \prod_j \Delta\mathrm{T}_{\text{est},j}$。当我们将预测模型$\mathrm{T}_{\text{pred},t}$（（即恒速预测)）应用于扫描的局部坐标系时，我们在机器人的全局参考系中执行ICP校正$\Delta\mathrm{T}_{\text{icp},t}$。这样做是出于效率原因，因为它允许我们在每次 ICP 迭代时仅变换源点 $\mathcal{S}$ 一次。由此，方程中使用的时间t处的局部姿态偏差$\Delta\mathrm{T}_t$。(5) 可以表示为

$$\Delta\mathsf{T}_t = (\mathsf{T}_{t-1}\mathsf{T}_{\text{pred},t})^{-1}\,\Delta\mathsf{T}_{\text{icp},t}\mathsf{T}_{t-1}\mathsf{T}_{\text{pred},t}. \tag{15}$$

ICP算法的标准终止标准是控制迭代次数。此外，大多数方法还具有基于解决方案的最小变化的进一步标准。相反，我们发现控制迭代次数并不能让算法总能找到好的解。因此，我们仅采用基于所应用的校正小于γ的终止标准，而不施加最大迭代次数。

最后，对点云$\mathcal{P}^*_{\text{merge}}$进行ICP校正，并将点集成到局部地图中。

F. Parameters

我们的实现取决于一小部分七个参数。全部显示在选项卡中。 I. 我们对所有实验使用相同的参数。大多数其他方法使用更大的参数集：MULLS [21] 有 107 个参数，SuMa [1] 有 49 个参数，CT-ICP [10] 在各自的配置文件中有 30 个参数。相比之下，我们的方法只有两个用于对应搜索的参数，四个用于地图表示和扫描子采样，一个用于 ICP 终止。请注意，扫描仪的最大范围是一个取决于所使用的特定传感器的值，因此，我们不将其视为系统参数。

Parameter	**Value**
Initial threshold τ_0	2 m
Min. deviation threshold $\delta_{\min}$	0.1 m
Max. points per voxel $N_{\max}$	20
Voxel size map v	$0.01\, r_{\max}$
Factor voxel size map merge α	0.5
Factor voxel size registration β	1.5
ICP convergence criterion γ	10^{-4}

TABLE I: All seven parameters of our approach.

However, for some scenarios, the value of $r_{\max}$ might also be adapted to the specific environment in which the system is operating, e.g., not considering far away measurements that are usually less accurate.

IV. Experimental Evaluation

This work provides a simple yet effective LiDAR odometry pipeline that comes with a small set of parameters. We present our experiments to show the capabilities of our method. The results of our experiments support our key claims, namely that our approach (i) is on par with more complex state-of-the-art odometry systems, (ii) can accurately compute the robot's odometry in a large variety of environments and motion profiles with the same system configuration, and (iii) provides an effective solution to motion distortion without relying on IMUs or wheel odometers.

A. Experimental Setup

We use numerous datasets and common evaluation methods. We start with the KITTI odometry dataset [12] to evaluate our system against state-of-the-art approaches to LiDAR odometry. To investigate how we perform in other autonomous driving datasets employing a different sensor, we evaluate our approach on the MulRan dataset [15]. Additionally, we show that our approach can be used in different scenarios, such as the one present in the NCLT dataset [5], a segway dataset, and the Newer College dataset [22] recorded using a handheld device. We also analyze our method's different components, such as the motion-compensation scheme and the adaptive threshold.

Please note that due to space limitations we omit to show the results of the trajectories and a detailed runtime evaluation in this manuscript but refer the reader to the official project page where all the plots and per-sequence evaluation on the runtime performances are available.[1]

B. Performance on the KITTI-Odometry Benchmark

This experiment evaluates the performance of different odometry pipelines on the popular KITTI benchmark dataset. Since most systems do not do motion compensation, we use the already compensated KITTI scans for a fair comparison and disable the motion compensation for our approach and CT-ICP [10] in this first analysis (the performance of the motion compensation module will be studied later in Sec. IV-D1). Tab. II exhibits how our system challenges most state-of-the-art systems, which are typically more sophisticated than our point-to-point ICP. Based on the official KITTI Benchmark,

[1]https://www.github.com/PRBonn/kiss-icp/tree/main/evaluation

	Method	**Seq. 00-10**	**Seq. 11-21**
SLAM	SuMa++ [1]	0.70	1.06
SLAM	MULLS [21]	0.52	-
SLAM	CT-ICP [10]	0.53	**0.59**
Odometry	IMLS-SLAM [11]	0.55	0.69
Odometry	MULLS [21]	0.55	0.65
Odometry	F-LOAM [33]	0.84	1.87
Odometry	SuMa [1]	0.80	1.39
Odometry	Ours	**0.50**	0.61

TABLE II: KITTI Benchmark results with motion compensated data. We report the average relative translational error in % [13]. We compare across SLAM methods employing pose-graph optimization for improved results *(top)* and odometry methods *(bottom)*. We omit the relative rotational error, but these results are available at https://www.cvlibs.net/datasets/kitti/eval_odometry.php

we rank second among the open-source approaches (behind CT-ICP [10]) and ninth among all submissions. This indicates that our comparably simple system still performs better than all the publicly available systems out there, except CT-ICP [10]. Note that CT-ICP is a complete SLAM system, and it uses loop closures to correct for the accumulated drift of the odometry estimation. We in contrast obtain our results using only open-loop registration without any loop closing.

C. Comparison to State-of-the-Art Systems on Other Datasets

We proceed to analyze the performance of our system on different datasets, scenarios, and types of robots. For that, we use the MulRan dataset [15], a handheld device [22], and a segway dataset [5]. Odometry pipelines typically deal with those challenging scenarios but employ IMUs [27] or a different system configuration [10]. Our system performs on par with state-of-the-art systems using the same parameter values for all experiments and datasets. For this experiment, we compare against state-of-the-art odometry systems, namely MULLS [21], SuMa [1], F-LOAM [33], and CT-ICP [10]. Note that we do not provide an evaluation of CT-ICP for the MulRan dataset since CT-ICP does not provide support for this dataset.

For the MulRan dataset [15], we test the systems under evaluation on all available public sequences. Since the dataset provides three similar runs for each sequence, we report the average number of each sequence in Tab. III. Our method outperforms all state-of-the-art approaches by a large margin in both relative and absolute error.

We use both available sequences to evaluate the Newer College dataset and achieve similar results on the short experiment compared to CT-ICP. For the long experiment, the performance gap can be explained by the additional loop closing module of CT-ICP, which is a complete SLAM system. For the NCLT dataset experiment, we use the sequence evaluated on the original work of CT-ICP. We could not reproduce the results reported in CT-ICP [10] and therefore report the results given in the original paper [10] in Tab. IV. We achieve similar results than CT-ICP. However, we observed errors in the GPS ground-truth poses and missing frames. Therefore, the numbers on NCLT should be taken with a grain of salt and rather provide an estimate of how the systems perform. We discourage using NCLT to evaluate odometry systems:

Parameter	Value
Initial threshold τ_0	2 m
Min. deviation threshold δ_{min}	0.1 m
Max. points per voxel N_{max}	20
Voxel size map v	0.01 r_{max}
Factor voxel size map merge α	0.5
Factor voxel size registration β	1.5
ICP convergence criterion γ	10^{-4}

表一：我们方法的所有七个参数。

然而，对于某些场景，r_{max}的值也可能适应系统运行的特定环境，例如，不考虑通常不太准确的远距离测量。

四. 实验评估

这项工作提供了一个简单而有效的 LiDAR 里程计管道，其中包含一小组参数。我们提出我们的实验来展示我们方法的能力。我们的实验结果支持了我们的关键主张，即我们的方法（i）与更复杂的最先进的里程计系统相当，（ii）可以在多种环境和具有相同系统配置的运动轨迹中准确计算机器人的里程计，以及（iii）提供运动失真的有效解决方案，而无需依赖IMU或车轮里程计。

A. Experimental Setup

我们使用大量数据集和常见的评估方法。我们从 KITTI 里程计数据集 [12] 开始，根据最先进的 LiDAR 里程计方法评估我们的系统。为了研究我们在使用不同传感器的其他自动驾驶数据集中的表现，我们在 MulRan 数据集上评估我们的方法 [15]。此外，我们表明我们的方法可以用于不同的场景，例如 NCLT 数据集 [5]、赛格威数据集和使用手持设备记录的 Newer College 数据集 [22] 中存在的场景。我们还分析了我们方法的不同组成部分，例如运动补偿方案和自适应阈值。

请注意，由于空间限制，我们省略了在本手稿中显示轨迹结果和详细的运行时评估，但请读者参考官方项目页面，其中提供了运行时性能的所有绘图和每个序列的评估。[1]

B. Performance on the KITTI-Odometry Benchmark

该实验评估了流行的 KITTI 基准数据集上不同里程计管道的性能。由于大多数系统不进行运动补偿，因此我们使用已经补偿的 KITTI 扫描进行公平比较，并在第一个分析中禁用我们的方法和 CT-ICP [10] 的运动补偿（运动补偿模块的性能将在第 IV-D1 节稍后研究）。标签。 II 展示了我们的系统如何挑战大多数最先进的系统，这些系统通常比我们的点对点 ICP 更复杂。基于官方 KITTI 基准测试，

	Method	Seq. 00-10	Seq. 11-21
MALS	SuMa++ [1]	0.70	1.06
	MULLS [21]	0.52	-
	CT-ICP [10]	0.53	**0.59**
yremodO	IMLS-SLAM [11]	0.55	0.69
	MULLS [21]	0.55	0.65
	F-LOAM [33]	0.84	1.87
	SuMa [1]	0.80	1.39
	Ours	**0.50**	0.61

表 II：带有运动补偿数据的 KITTI 基准测试结果。我们以百分比报告平均相对平移误差[13]。我们比较了采用位姿图优化来改进结果的 SLAM 方法 *(top)* 和里程计方法 *(bottom)*。我们省略了相对旋转误差，但这些结果可在 https://www.cvlibs.net/datasets/kitti/eval odometry.php 获得

我们在开源方法中排名第二（仅次于 CT-ICP [10]），在所有提交的方法中排名第九。这表明我们相对简单的系统仍然比除 CT-ICP 之外的所有公开可用系统表现更好 [10]。请注意，CT-ICP 是一个完整的 SLAM 系统，它使用闭环来校正里程计估计的累积漂移。相比之下，我们仅使用开环配准而无需任何闭环来获得结果。

C. Comparison to State-of-the-Art Systems on Other Datasets

我们继续分析我们的系统在不同数据集、场景和机器人类型上的性能。为此，我们使用 MulRan 数据集 [15]、手持设备 [22] 和 segway 数据集 [5]。里程计管道通常处理那些具有挑战性的场景，但采用 IMU [27] 或不同的系统配置 [10]。我们的系统的性能与最先进的系统相当，所有实验和数据集使用相同的参数值。在本实验中，我们与最先进的里程计系统进行比较，即 MULLS [21]、SuMa [1]、F-LOAM [33] 和 CT-ICP [10]。请注意，我们不提供 MulRan 数据集的 CT-ICP 评估，因为 CT-ICP 不提供对此数据集的支持。

对于 MulRan 数据集 [15]，我们在所有可用的公共序列上测试评估中的系统。由于数据集为每个序列提供了三个相似的运行，因此我们在选项卡中报告了每个序列的平均数。三．我们的方法在相对误差和绝对误差方面都大大优于所有最先进的方法。

我们使用两个可用序列来评估 Newer College 数据集，并在短期实验中获得与 CT-ICP 类似的结果。对于长时间的实验，性能差距可以通过 CT-ICP 的附加闭环模块来解释，CT-ICP 是一个完整的 SLAM 系统。对于 NCLT 数据集实验，我们使用在 CT-ICP 原始工作中评估的序列。我们无法重现 CT-ICP [10] 中报告的结果，因此报告表 1 中原始论文 [10] 中给出的结果。四．我们取得了与 CT-ICP 相似的结果。然而，我们观察到 GPS 地面实况姿势存在错误以及丢失帧。因此，NCLT 上的数字应该持保留态度，而是提供对系统性能的估计。我们不鼓励使用 NCLT 来评估里程系统：

[1]https://www.github.com/PRBonn/kiss-icp/tree/main/evaluation

Sequence	Method	Avg. tra.	Avg. rot.	ATE tra.	ATE rot.
KAIST	MULLS [21]	2.94	0.86	37.24	0.11
	SuMa [1]	5.59	1.73	43.61	0.14
	F-LOAM [33]	3.43	0.99	46.17	0.15
	Ours	**2.28**	**0.68**	**17.40**	**0.06**
DCC	MULLS [21]	2.96	0.98	38.35	0.12
	SuMa [1]	5.20	1.71	36.22	0.11
	F-LOAM [33]	3.83	1.14	42.70	0.13
	Ours	**2.34**	**0.64**	**15.16**	**0.05**
Riverside	MULLS [21]	5.42	2.21	91.16	0.16
	SuMa [1]	13.86	2.13	227.24	0.38
	F-LOAM [33]	5.47	1.18	138.09	0.22
	Ours	**2.89**	**0.64**	**49.02**	**0.08**
Sejong*	MULLS [21]	5.93	0.84	2151.00	0.49
	F-LOAM [33]	7.87	1.20	3448.97	0.82
	Ours	**4.69**	**0.70**	**1369.54**	**0.33**

TABLE III: Quantitative results on the MulRan dataset [15]. We report the relative translational error and the relative rotational error using the KITTI [13] metrics. Additionally, we show the absolute trajectory error for translation in m and for rotation in rad.

Method	NCD 01-short	NCD 02-long	NCLT 2012-01-8
MULLS [21]	0.82	1.23	-
F-LOAM [33]	2.02	fails	-
CT-ICP [10]	**0.48**	**0.58**	**1.17**
Ours	0.51	0.96	1.27

TABLE IV: Quantitative results for Newer College and NCLT. We report the relative translational error in % [13].

misalignments in the ground truth poses, missing frames, and inconsistencies in the data make the evaluation of odometry systems on such a dataset not a good evaluation tool from our perspective. However, we provide the results for completeness.

We show qualitative results in Fig. 1 generated using our KISS-ICP poses. Using a single system configuration, we can produce consistent maps on different sensor setups (Velodyne/Ouster vs. Livox) and different motion profiles (car, drone, segway, handheld) with the same parameters.

D. Ablation Studies

To understand how each component of our system impacts the odometry performance, we conduct ablation studies on the different components of our approach, namely, the motion compensation scheme and the adaptive threshold. To carry out these studies, we use the KITTI odometry dataset [12] as it is probably the best-known one.

1) Motion Compensation: To assess the impact of our motion compensation scheme, we utilize the raw LiDAR point clouds without any compensation applied. Note that the KITTI odometry benchmark point cloud data [12] is already compensated and, therefore, cannot be used for this study. Thus, we use the KITTI raw dataset [13]. We present the results in a familiar fashion, selecting only the sequences that correspond to the ones on the motion-compensated datasets [12]. As we can see in Tab. V, our motion compensation scheme can produce state-of-the-art results and is on par with substantially more sophisticated and thus complex compensation techniques such as the one introduced by CT-ICP [10]. Additionally, we study how our system performs without applying motion compensation, as shown in Tab. V. We also evaluate the performance of our constant velocity model for motion compensation. To assess this, we compare the same compensation strategy but replace the velocity estimation with sensor data taken by the IMU. As seen in the results, our velocity estimation is on par or even slightly better with the IMU.

Method	Avg. tra	Avg. rot	Avg. freq.
MULLS [21]	1.41	-	12 Hz
IMLS-SLAM [11]	0.71	-	1 Hz
CT-ICP [10]	0.55	-	15 Hz
Ours without deskewing	0.91	0.27	51 Hz
Ours + Deskewing (IMU)	0.51	0.19	38 Hz
Ours + Deskewing (CV)	**0.49**	0.16	38 Hz

TABLE V: Results of evaluating different state-of-the-art systems on KITTI-raw dataset (without motion compensation). We report the relative translational error and the relative rotational error using the KITTI [13] metrics. Additionally, we report the runtime operation of the systems being in consideration for this experiment.

Dataset	Data-Association Threshold τ				
	0.3 m	0.5 m	1.0 m	2.0 m	Ours
KITTI Seq. 00	0.54	**0.51**	0.53	0.55	**0.51**
KITTI Seq. 04	0.39	0.41	0.37	0.39	**0.36**
KITTI Avg. Seq. 00-10	0.53	0.51	0.51	0.53	**0.50**

TABLE VI: Comparison of different fixed thresholds vs. our proposed adaptive threshold on the KITTI dataset. We report the relative translational error in % [13].

Besides the fact that CT-ICP's elastic formulation yields good results, our much simpler approach produces even better results. This result shows that the constant velocity model employed in our approach for compensating motion distortion is sufficient to cope with the slight reduction in performance when no compensation is applied. Consequently, we believe that more sophisticated techniques are unnecessary for *most* robotic odometry estimation.

2) Adaptive Data-Association Threshold: We finally evaluate how the adaptive threshold τ_t impacts the performance of our system by comparing it to a different set of fixed thresholds commonly used in open-source systems. To conduct this experiment, we identify the two KITTI sequences with the largest (00) and the smallest (04) average acceleration indicating different motion profiles. As we can see in Tab. VI, the best fixed threshold for sequence 00 is 0.5 m and 1.0 m for sequence 04. This means that a fixed threshold has to be tuned depending on the motion profile and thus to the dataset to achieve top performance. In contrast, our adaptive algorithm exploits the motion profile to estimate the threshold online, which results in on-par or better performance without the need to find a new fixed threshold for each sequence. Finally, our proposed adaptive threshold strategy achieves the best average result on the KITTI training sequences.

Please note that all the experiments from this ablation study use the robust kernel. For space reasons, we omitted the results of the evaluation of our system when no kernel is employed and only report the results averaged over the sequences. Not using the kernel produces 0.67% for the translational error and 0.25% for the rotational error [13].

Sequence	Method	Avg. tra.	Avg. rot.	ATE tra.	ATE rot.
KAIST	MULLS [21]	2.94	0.86	37.24	0.11
	SuMa [1]	5.59	1.73	43.61	0.14
	F-LOAM [33]	3.43	0.99	46.17	0.15
	Ours	**2.28**	**0.68**	**17.40**	**0.06**
DCC	MULLS [21]	2.96	0.98	38.35	0.12
	SuMa [1]	5.20	1.71	36.22	0.11
	F-LOAM [33]	3.83	1.14	42.70	0.13
	Ours	**2.34**	**0.64**	**15.16**	**0.05**
Riverside	MULLS [21]	5.42	2.21	91.16	0.16
	SuMa [1]	13.86	2.13	227.24	0.38
	F-LOAM [33]	5.47	1.18	138.09	0.22
	Ours	**2.89**	**0.64**	**49.02**	**0.08**
Sejong*	MULLS [21]	5.93	0.84	2151.00	0.49
	F-LOAM [33]	7.87	1.20	3448.97	0.82
	Ours	**4.69**	**0.70**	**1369.54**	**0.33**

表 III：MulRan 数据集的定量结果 [15]。我们使用 KITTI [13] 指标报告相对平移误差和相对旋转误差。此外，我们还显示了平移的绝对轨迹误差（以 m 为单位）和旋转的绝对轨迹误差（以 rad 为单位）。

Method	NCD 01-short	NCD 02-long	NCLT 2012-01-8
MULLS [21]	0.82	1.23	-
F-LOAM [33]	2.02	fails	-
CT-ICP [10]	**0.48**	**0.58**	**1.17**
Ours	0.51	0.96	1.27

表 IV：Newer College 和 NCLT 的定量结果。我们以百分比报告相对平移误差[13]。

从我们的角度来看，地面真实姿势的错位、丢失的帧以及数据的不一致使得在这样的数据集上评估里程计系统并不是一个好的评估工具。但是，我们提供结果是为了完整性。

我们在图 1 中展示了使用 KISS-ICP 姿势生成的定性结果。使用单一系统配置，我们可以在不同的传感器设置（Velodyne/Ouster 与 Livox）和不同的运动配置文件（汽车、无人机、赛格威、手持设备）上使用相同的参数生成一致的地图。

D. Ablation Studies

为了了解我们系统的每个组件如何影响里程计性能，我们对我们方法的不同组件（即运动补偿方案和自适应阈值）进行了消融研究。为了进行这些研究，我们使用 KITTI 里程计数据集 [12]，因为它可能是最著名的数据集。

1) Motion Compensation: 为了评估我们的运动补偿方案的影响，我们利用原始 LiDAR 点云，而不应用任何补偿。请注意，KITTI 里程计基准点云数据 [12] 已经经过补偿，因此不能用于本研究。因此，我们使用 KITTI 原始数据集 [13]。我们以熟悉的方式呈现结果，仅选择与运动补偿数据集上的序列相对应的序列[12]。正如我们在选项卡中看到的。 V，我们的运动补偿方案可以产生最先进的结果，并且与更复杂的补偿技术（例如 CT-ICP [10] 引入的技术）相当。此外，我们研究

Method	Avg. tra	Avg. rot	Avg. freq.
MULLS [21]	1.41	-	12 Hz
IMLS-SLAM [11]	0.71	-	1 Hz
CT-ICP [10]	0.55	-	15 Hz
Ours without deskewing	0.91	0.27	51 Hz
Ours + Deskewing (IMU)	0.51	0.19	38 Hz
Ours + Deskewing (CV)	**0.49**	0.16	38 Hz

表 V：在 KITTI 原始数据集上评估不同最先进系统的结果（无运动补偿）。我们使用 KITTI [13] 指标报告相对平移误差和相对旋转误差。此外，我们报告了本实验考虑的系统运行时操作。

Dataset	Data-Association Threshold τ				
	0.3 m	0.5 m	1.0 m	2.0 m	Ours
KITTI Seq. 00	0.54	**0.51**	0.53	0.55	**0.51**
KITTI Seq. 04	0.39	0.41	0.37	0.39	**0.36**
KITTI Avg. Seq. 00-10	0.53	0.51	0.51	0.53	**0.50**

表 VI：不同固定阈值与我们在 KITTI 数据集上提出的自适应阈值的比较。我们以百分比报告相对平移误差[13]。

我们的系统在不应用运动补偿的情况下如何执行，如表 1 所示。 V. 我们还评估了用于运动补偿的恒速模型的性能。为了评估这一点，我们比较了相同的补偿策略，但用 IMU 获取的传感器数据替换了速度估计。从结果中可以看出，我们的速度估计与 IMU 相当甚至稍好一些。

除了 CT-ICP 的弹性公式可以产生良好的结果之外，我们更简单的方法也可以产生更好的结果。该结果表明，我们的方法中用于补偿运动失真的恒速模型足以应对不应用补偿时性能的轻微下降。因此，我们认为 *most* 机器人里程计估计不需要更复杂的技术。

2) Adaptive Data-Association Threshold: 最后，我们通过将自适应阈值 τ_t 与开源系统中常用的一组不同的固定阈值进行比较来评估它如何影响我们系统的性能。为了进行此实验，我们识别了具有最大 (00) 和最小 (04) 平均加速度的两个 KITTI 序列，表明不同的运动曲线。正如我们在选项卡中看到的。 VI，序列 00 的最佳固定阈值是 0.5 m，序列 04 的最佳固定阈值是 1.0 m。这意味着必须根据运动轮廓以及数据集调整固定阈值才能实现最佳性能。相比之下，我们的自适应算法利用运动轮廓来在线估计阈值，从而获得同等或更好的性能，而无需为每个序列找到新的固定阈值。最后，我们提出的自适应阈值策略在 KITTI 训练序列上实现了最佳平均结果。

请注意，该消融研究中的所有实验都使用稳健的内核。由于篇幅原因，我们省略了未使用内核时系统的评估结果，仅报告序列上的平均结果。不使用内核会产生 0.67% 的平移误差和 0.25% 的旋转误差 [13]。

V. Conclusion

This paper presents a simple yet highly effective approach to LiDAR odometry and shows that point-to-point ICP works very well – when used properly. Our approach operates solely on point clouds and does not require an IMU, even when dealing with high-frequency driving profiles. Our approach exploits the classical point-to-point ICP to build a generic odometry system that can be employed in different challenging environments, such as highway runs, handheld devices, segways, and drones. Moreover, the system can be used with different range-sensing technologies and scanning patterns. We only assume that point clouds are generated sequentially as the robot moves through the environment. We implemented and evaluated our approach on different datasets, provided comparisons to other existing techniques, supported all claims made in this paper, and released our code. The experiments suggest that our approach is on par with substantially more sophisticated state-of-the-art LiDAR odometry systems but relies only on a few parameters, and performs well on various datasets under different conditions with the same parameter set. Finally, our system operates faster than the sensor frame rate in all presented datasets. We believe this work will be a new baseline for future sensor odometry systems and a solid, high-performance starting point for future approaches. Our open-source code is robust and simple, easy to extend, and performs well, pushing the state-of-the-art LiDAR odometry to its limits and challenging most sophisticated systems.

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五、结论

本文提出了一种简单而高效的 LiDAR 测距方法，并表明如果使用得当，点对点 ICP 效果非常好。我们的方法仅在点云上运行，即使在处理高频驾驶曲线时也不需要 IMU。我们的方法利用经典的点对点 ICP 来构建通用里程计系统，该系统可用于不同的挑战性环境，例如高速公路、手持设备、赛格威和无人机。此外，该系统可以与不同的范围传感技术和扫描模式一起使用。我们仅假设当机器人在环境中移动时点云是顺序生成的。我们在不同的数据集上实施和评估了我们的方法，提供了与其他现有技术的比较，支持本文中提出的所有主张，并发布了我们的代码。实验表明，我们的方法与更复杂的最先进的激光雷达里程计系统相当，但仅依赖于几个参数，并且在不同条件下具有相同参数集的各种数据集上表现良好。最后，我们的系统在所有呈现的数据集中都比传感器帧速率运行得更快。我们相信这项工作将成为未来传感器里程计系统的新基准，也是未来方法的坚实、高性能的起点。我们的开源代码强大且简单，易于扩展且性能良好，将最先进的 LiDAR 里程计推向极限，并对最复杂的系统提出挑战。

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